

A Written de Sitter–Unruh Modified-Inertia Action (v3)

The frame field is passive, the covariant theory is causal and ghost-free at two-point and Cassini-safe, the Einstein-aether strong-coupling wall is mis-applied, and gravitational lensing is viable but tuned

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Abstract

The de Sitter–Unruh reading of the MOND acceleration scale, $a_0 = cH_\Lambda/Z = c^2\sqrt{(\Lambda/32\pi)} = 9.36 \times 10^{-11} \text{ m s}^{-2}$, posits that the scale is a modified-**inertia** effect of the cosmic (dark-energy) horizon rather than a modification of gravity. This paper writes the explicit nonlocal modified-inertia action and carries its covariant analysis to a definite boundary, reasoning throughout from the framework’s own premises (a passive frame, with the MOND in the matter sector) rather than the standard aether/MOND lens. Results, each with reproducible verification (both a_0 footings): (1) the **worldline/matter sector is machine-verified** ($v(y)=\sqrt{(1+1/y)}$ to 3×10^{-13} , Newton, deep-MOND, baryonic Tully–Fisher, the dS-Unruh-forced $\sqrt{2}$ -DC-weight external-field kernel, exact ghost-freedom). (2) The modified-**inertia** realization **evades the Cassini quadrupole** by ~ 7 orders (deep-Newtonian $v-1 \approx 7 \times 10^{-7}$), where the modified-gravity realization fails at $+6$ to $+14\sigma$. (3) Promoting the frame field covariantly, **MOND does not force modified gravity** — the matter source is $l=0$ (soaked by the unit-timelike multiplier), evading the Bianchi shear lock. (4) The mixed matter–aether **two-point propagator is causal and ghost-free in closed form** (retarded construction; Källén–Lehmann spectral positivity across the whole cut; principal symbol = the GR light-cone). (5, **correcting v2**) The Einstein-aether **strong-coupling wall does not wall the framework**: that wall is a property of a *propagating* aether mode whose kinetic norm vanishes, but the framework’s frame is **passive** (its own theory: an algebraic constitutive law, an SME background, the cosmic rest frame), with zero propagating aether modes — so the wall’s object is undefined for it. The wall binds only the rejected AeST/khronon *modified-gravity* limb. (6) Gravitational **lensing is viable but fine-tuned**: light is massless, so lensing must come from the ghost-condensate dark sector, which reproduces the observed lensing=dynamics relation only if placed by hand to equal the MI dynamical excess (per-galaxy) — not excluded, not predicted. **Open**: an all-orders certifying well-posedness theorem for the non-entire branch-cut class, the interacting four-point vertex, and a first-principles condensate profile law. The MOND sign remains a **postulate** and a_0 ’s value is underived. An effective one-scale completion carried to a sharp, named boundary — **not a finished theory and not a theory of everything**.

1. Setting and scope

The framework reads a_0 as the acceleration at which a body’s inertial response to the de Sitter vacuum departs from Newtonian, with the Deser–Levin temperature $T(a) = (\hbar/2\pi k_B c) \sqrt{(a^2 + (cH_\Lambda)^2)}$ a horizon floor and the interpolation $g_{\text{obs}} = \sqrt{(g_{\text{bar}}^2 + g_{\text{bar}} a_0)}$, $v(y)=\sqrt{(1+1/y)}$. The earlier theory-of-everything and Standard-Model claims were publicly retracted (2026-06-23) and are not reasserted; this is a one-scale effective-theory construction with named inputs,

each labeled **derived**, **postulated**, or **open**. Crucially — and this is the correction to v2 — the framework is *modified inertia*: the MOND lives in the **matter** sector, and the frame field u^μ is a **passive**, horizon-anchored reference (the framework’s own paper: an *algebraic constitutive law* $g_{\text{bar}}=G(a)$ with no frame equation of motion; an SME gravity-sector *background*; the Cassini-constrained cosmic rest frame). It is **not** AeST/Einstein-aether, where a dynamical aether *carries* the MOND — the theory that fails Cassini.

2. The action

With signature $(-+++)$, $S[x,g,u] = S_{\text{EH}}[g] + S_u[g,u] + S_{\text{matter}}[x,g,u]$: host GR unmodified; a unit-timelike frame constraint $S_u = -\int \sqrt{-g} (\lambda/2)(u^\mu \mu_\mu + 1)$ (no aether kinetic term); and the modified-inertia content in the matter kinetic sector,

$$S_{\text{matter}} = -\frac{1}{2} \int d^4x \sqrt{-g} \rho_m [s u^\mu K(\square_u/a_0^2) u_\mu], \quad K(z) = \frac{\sqrt{1+4z}-1}{2\sqrt{z}}, \quad \square_u f = u^a \nabla_a (u^b \nabla_b f).$$

K is non-polynomial with a branch cut and a single healthy pole (residue $+1$). The equivalent Galley doubled-worldline form uses the conservative gradient $G(a)^{\mu=a} \mu_\mu \text{fw}(|a|/a_0)$, $\mu_\mu \text{fw}$ the exact inverse of $v(y)=\sqrt{(1+1/y)}$; the external-field kernel $\theta(y)=\theta_0/(1+(\theta_0-1)y^2)$, $\theta_0=\sqrt{2}$, has its shape forced by the de Sitter Wightman function. Here $s=-1$ is the **MOND-sign postulate**, $a_0=cH_\Lambda/Z=9.36 \times 10^{-11}$ (canonical), 1.13×10^{-10} (alternate footing).

3. Verified: the worldline sector and the Cassini evasion

Machine-checked, both footings: the circular-orbit reduction reproduces $v(y)=\sqrt{(1+1/y)}$ to 3.35×10^{-13} ; Newton, deep-MOND, and baryonic Tully–Fisher $v^4=GM/a_0$ (exact); the form-factor $\leftrightarrow$ RAR identity $g_{\text{obs}}=\sqrt{(g_{\text{bar}}^2+g_{\text{bar}} a_0)}$; the $\sqrt{2}$ -DC-weight external-field kernel; and ghost-freedom (single pole $+1$ vs the Ostrogradsky ghost of the local truncation). **Cassini**: the anisotropic galactic external field ($a_{\text{ext}}=2.29 a_0$) enters only Saturn’s inertial response, deep-Newton-suppressed by $v-1 \approx 7 \times 10^{-7}$; the $l=2$ quadrupole is $\sim 7.4 \times 10^{-34} \text{ s}^{-2} \sim 7$ orders below the 5.2×10^{-27} ceiling. The modified-gravity realization instead fails at $+6$ to $+14\sigma$. This is Milgrom’s [2009] statement that the inner-solar-system quadrupole is an AQUAL/QUMOND effect absent in modified inertia, realized explicitly.

4. The covariant completion, on the framework’s own terms

Because the frame is passive and the MOND is in the matter sector, the covariant question is *not* whether a dynamical aether is strongly coupled — it is whether the passive-frame + nonlocal-K matter system is well-posed. Four results:

(a) MOND does not force modified gravity. Varying S_{matter} , the source it adds to u^μ ’s field equation is $J^\mu = -\rho_m s k u^\mu$: $l=0$, parallel to u^μ , soaked by the multiplier λ . It never populates the $l=2$ traceless-shear channel whose divergence is the Bianchi lock that forces AeST’s Cassini quadrupole. So the covariant lift does not collapse to modified gravity; a passive frame carries at most $Q_2^\wedge u \lesssim 7 \times 10^{-35} \text{ s}^{-2}$ over an open, ghost-free region.

(b) The mixed two-point propagator is causal and ghost-free in closed form. The dynamical- u variation resums to $Q(z)=K+2zK'=2\sqrt{z}/\sqrt{(1+4z)}$; the retarded memory kernel has strictly

causal support (an advanced control correctly flips it), Titchmarsh UHP-analyticity holds, and Källén–Lehmann spectral positivity $\rho(s) \geq 0$ across the whole cut is verified in closed form (a control correctly flags a known Ostrogradsky ghost as $\rho < 0$). The branch cut is a healthy radiation continuum, not a ghost.

(c) The principal symbol is the GR light-cone, $D(w)/w = 1 + i a_0/2w - a_0^2/8w^2 + \dots$, the nonlocal correction $O(a_0/w)$ UV-subdominant and IR-gapped — hyperbolic, no secular runaway. So the passive-frame theory is **principal-symbol-well-posed**.

(d) The Einstein-aether strong-coupling wall is mis-applied (correcting v2). That wall is, by construction, a property of a *propagating* aether/khronon mode whose canonical kinetic norm vanishes ($M_{sc} \sim \sqrt{N} M_{Pl}$; the mode speeds carry the vanishing norm in the denominator) [Blas–Pujolàs–Sibiryakov 2009; Gong et al. 2018]. The framework’s frame is passive — zero such modes — so the wall’s central object is *undefined*, not violated. It binds only the rejected AeST/khronon modified-gravity limb (which fails Cassini at $+6-14\sigma$). v2’s “the covariant route is walled by strong coupling” applied that dynamical-aether wall to the passive-frame realization and is hereby corrected.

What is open. There is no off-the-shelf *certifying* all-orders well-posedness theorem for the non-entire branch-cut (Källén–Lehmann/Herglotz) class — positivity + analyticity are passed checks, not a theorem (the proven literature covers only *entire* form factors). And the interacting four-point vertex plus nonlinear back-reaction (Box_u is built from u, so “linear on a fixed passive-u background” is the controlled slice) is not closed in one pass.

5. Gravitational lensing: viable but fine-tuned

In modified inertia the modification is to how massive bodies respond; **light is massless and has no inertia to modify**, so photons follow GR null geodesics of real mass — baryons plus the ghost-condensate dark sector. The observed weak-lensing RAR matches the dynamical RAR [Brouwer et al. 2021; Misteale & McGaugh 2024], which *requires* the condensate to supply exactly the MI dynamical excess, $M_{cond}(r) = M_{bar}(v(y)-1)$. Nothing in the framework forces this: the condensate amount is a free shift-charge integration constant ($dI_0/da_0=0$), and generic clustering profiles miss the required shape by 0.12–0.55 dex — only a hand-placed profile matches. Worse, a fixed cosmic I_0 gives a constant M_{cond}/M_{bar} while the MI excess runs as $\sqrt{M_{bar}}$ across the galaxy population, so the tuning is *per-galaxy*. This is **not a tension** ($M_{cond}=M_{bar}(v-1)$ is positive-density, monotonic, and realizable — the framework is not excluded) and **not a prediction** (it is a fit). Verdict: viable but fine-tuned, the location of generic two-sector dark theories. The genuinely distinctive modified-inertia signature lives in the *dynamical* sector (the MG-impossible non-adiabatic velocity-dispersion hysteresis), not in lensing.

6. Honest standing

Reasoning from the framework’s own passive-frame, MOND-in-matter premise: the worldline sector is written and verified; the modified-inertia realization evades Cassini; MOND does not force modified gravity; the mixed two-point propagator is causal, ghost-free (Källén–Lehmann), and principal-symbol-well-posed; and the Einstein-aether strong-coupling wall is a mis-application that does not bind the framework (it binds only the rejected modified-gravity

limb). Gravitational lensing is viable but fine-tuned (per-galaxy), not solved and not a kill. The completion is therefore **not** the modified-gravity limb (Cassini-failing) and **not yet** a finished field theory — it is a well-posed worldline theory whose passive-frame covariantization is causal, ghost-free at two-point, and principal-symbol-well-posed, with three honest open edges: a certifying all-orders theorem for the non-entire branch-cut class, the interacting four-point vertex, and a first-principles ghost-condensate profile law. The MOND sign stays postulated and a_0 's value underived. No completeness or theory-of-everything claim is made.

7. Related work

The construction sits between the relativistic modified-gravity realizations of MOND — AeST [Skordis & Złośnik 2021], the Blanchet–Skordis khronon [2025], the nonlocal mimetic model of Deffayet & Woodard [2026] — and the modified-inertia program of Milgrom [1999, 2009]. The covariant frame sector is Einstein-aether [Jacobson & Mattingly 2001], its PPN structure [Foster & Jacobson 2006], its GW170817 constraint [Gong et al. 2018], and its strong-coupling corner [Blas, Pujolàs & Sibiryakov 2009]; the nonlocal well-posedness class is Källén–Lehmann/Herglotz–Nevanlinna (passive-causal), distinct from the entire-function class [Biswas et al.; Barnaby & Kamran]. The doubled-worldline formalism is Galley [2013]; the lensing constraint is Brouwer et al. [2021] and Mistele & McGaugh [2024].

8. Reproducibility

Committed scripts (repository `real_research/reviews/`): the worldline sector and Cassini evasion (`cassini_mi_evasion_2026/`); the covariant kinetic wins (`mi_kinetic_completion_2026/`); the mixed-propagator causality + Källén–Lehmann positivity (`mi_propagator_2026/`); the framework-first correction of the strong-coupling wall (`mi_strongcoupling_framework_2026/`); the all-orders Cauchy checks (`mi_cauchy_wellposed_2026/`); and the lensing fine-tuning analysis (`mi_lensing_2026/`). Sign-wall inputs: `ACTIVE_KERNEL_SIGNTHEOREM_2026-06.md`, `FOURTH_ORDER_SIGN_WALL_2026-07.md`; trichotomy: `COVARIANT_MI_COMPLETION_2026-06.md`.

References

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12. Companion (this program): sign wall, Zenodo 10.5281/zenodo.21184373; covariant-MI quadri-chotomy, 10.5281/zenodo.21148494; written MI action v_1/v_2 , 10.5281/zenodo.21253645 / 21260830.

Both a_0 footings throughout. Each claim labeled derived / postulated / open; the MOND sign is a postulate; the Einstein-aether wall is corrected as a mis-application; lensing is viable-but-tuned. No completeness or theory-of-everything claim.